EGELAND AUTO
Eigeland 50
Tel: 91131664 bernt.christian@egeland.io Org: NO911782162

# INVOICE

2026-1033 Feb 12, 2026 Due: Feb 26, 2026

**BILL TO**
**Frode Berntsen**

**VEHICLE**
**2017 BMW X5 (F15) xDrive**
Plate: BT74876
Mileage: 86,645 km

**SERVICE**
**Turbo, Eksos Manifoild Bolter, Klemme**
Type: repair
Tech: Bernt Christian Egeland

## Parts

| Part # | Description | Qty | Unit Price | Total |
|---|---|---|---|---|
| - | Turbo | 1 | 9 036,30 kr | **9 036,30 kr** |
| - | Eksos Manifoild Bolter | 2 | 40,80 kr | **81,60 kr** |
| - | Klemme | 1 | 230,00 kr | **230,00 kr** |
| - | Oljerør Turbo | 1 | 270,00 kr | **270,00 kr** |
| - | BMW Longlife Frostvæske | 2 | 250,00 kr | **500,00 kr** |
| - | Innsug temp/trykk sensor | 1 | 2 050,00 kr | **2 050,00 kr** |
| - | Olje | 5 | 320,00 kr | **1 600,00 kr** |

## Labor

| Description | Hours | Rate | Total |
|---|---|---|---|
| Turboen ble demontert og byttet. Den nye turboen var ikke helt identisk med originalen, noe som medførte ekstra arbeid. Blant annet måtte de nedre festepunktene skimmes ut med noen millimeter for å tilpasses. | 18 | 350,00 kr/hr | **6 300,00 kr** |
| Eksos feste til katalysator knakk under demontering, veldig rusta. Sveiset på nye bolter. | 1 | 350,00 kr/hr | **350,00 kr** |
| Oljeskift med filter. Resatt oljeservice. | 0.75 | 350,00 kr/hr | **262,50 kr** |
| Feilsøkt innsug temp som er 97.5 grader konstant. Dette gjør at turbo går i nødmodus med redusert turbotrykk. Målt sensor, den er på utsiden av spesifikasjoner fra BMW. | 2 | 350,00 kr/hr | **700,00 kr** |
| Kalibrert turbo wastegate ventil med BMW programvare. | 0.3 | 350,00 kr/hr | **105,00 kr** |
| Smørt opp hjulbolter høyre fremme, veldig rusta bør gjenges opp på nytt. | 0.1 | 350,00 kr/hr | **35,00 kr** |
| Fyllt frostvæske og målt til -30 Grader. | 0 | 0,00 kr/hr | **0,00 kr** |
| Fyllt luft i alle dekk, Høyre Fremme var veldig lite luft. Resatt Dekktrykk Overvåkning | 0 | 350,00 kr/hr | **0,00 kr** |

Parts 13 767,90 kr

| | |
|---|---|
| Labor | 7 752,50 kr |
| Subtotal | 21 520,40 kr |
| **Total** | **21 520,40 kr** |

**Payments Received**

| | |
|---|---|
| Feb 16, 2026 (other) | 9 470,40kr |
| Feb 8, 2026 (bank_transfer) | 2 050,00kr |
| Feb 8, 2026 (bank_transfer) | 10 000,00kr |

# PAID IN FULL

**NOTES**

**Anbefalinger / Notater:**

- Bytt motorolje på 15000.- og ikke 31000km intervall.
- 2 Bolter knakk til deksel i motorrom, må borres ut og gjenge opp. Ikke kritisk, men greit å få gjort på ett tidspunkt.
- Feilkode på startbatteriet: Aldret batteri. Ikke kritisk. Se vedlagt diagnose rapport.
- Feilkode på høyspentbatteri, celleblokk: Økt utlading av en celle. Bør følges med på fremover. Kan tyde på en dårlig battericelle i høyspentbatteriet. Se vedlagt diagnose rapport.

**TIL KONTO / BANK ACCOUNT**

**2850.10.39472**

**ATTACHED DOCUMENTS**

post-repair.pdf (see appended pages)
pre-repair.pdf (see appended pages)

View your portal: https://app.torqvoice.com/portal/egeland_auto

Payment Terms: Net 14 Days

## Service Images

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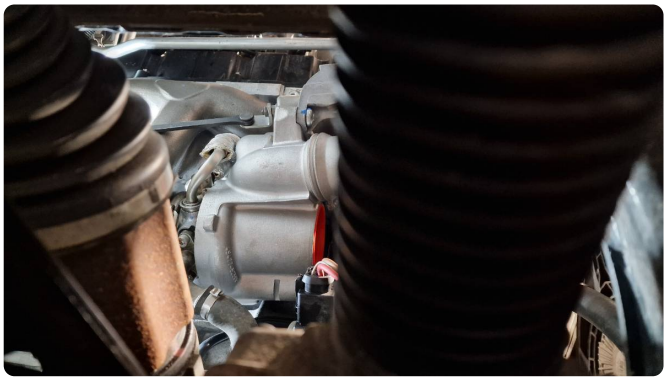

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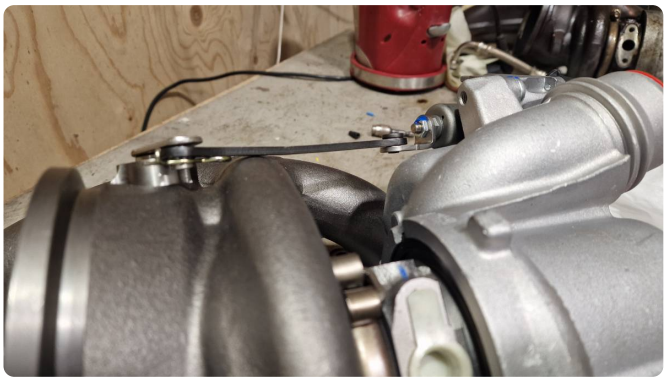

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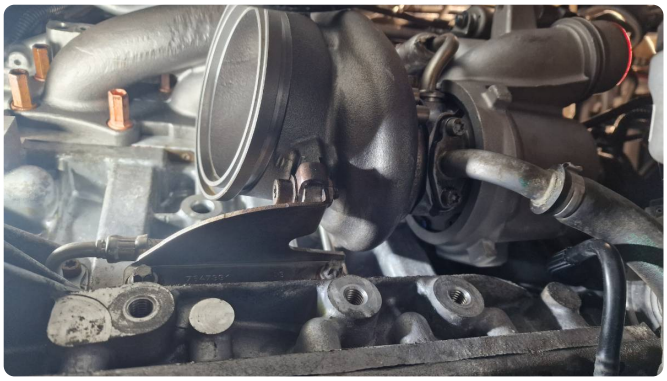

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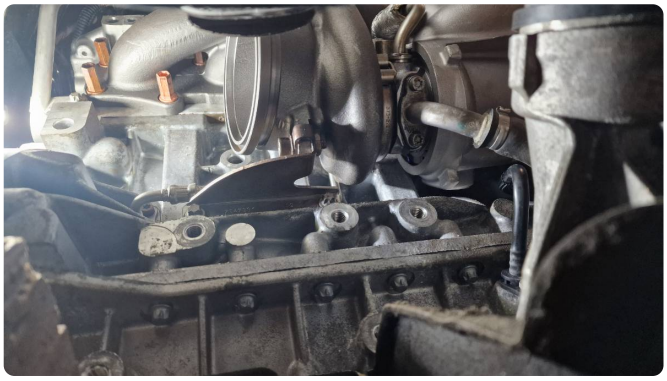

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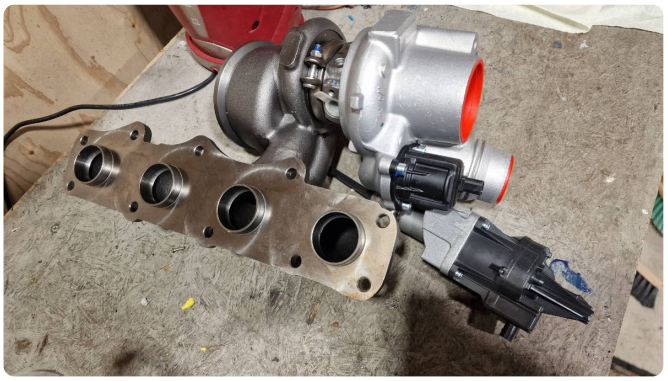

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# All System Diagnostic Report

The Report is created by THINKCAR

## Repair Shop

Shop Name:Egeland Diagnostic
Address:Eigeland 50
Telephone:N/A
Sn:988569501368
Time:2026-02-12 19:32:43

## Vehicle Information

Year:2017.08
Make:BMW
Model:X5 xDrive40e iPerfor
VIN:WBAKT010XH0Y43663
Odometer:86645 miles
Vehicle Software Version:V10.30

## Inspection Result

EME2 (Electrical Machine Electronics 2) 1 problems exist

1.2228C8 Starter Unit, Auxiliary Battery: Aged Auxiliary Battery Permanent

The following systems are OK:
1.SRS (Supplemental Inflatable Restraint System - AIRBAG/SGM-SIM)
2.TRSVC (All Round View Camera)
3.SME (Battery Management Electronics)
4.HC/SWW (Lane Change Warning)/SRR_HR (Rear Radar Sensor Near Range Right)
5.HKFM/HKL (Automatic Luggage Compartment Lid Actuation)
6.ZGM (Central Gateway Module)
7.ECM (Engine Control Module - DME/DDE)
8.LIM (Charging Interface Module)
9.EKPS (Electronic Fuel Pump)/RLM_1L (Tail Light Module, Outer Left)
10.TCM (Transmission Control Module - EGS)
11.LMV/VTG (Transfer Case)
12.ICM (Integrated Chassis Management)/RLM_2L (Tail Light Module, Inner Left)
13.TFM (Pressurized Fuel Tank Control Module)
14.ACC (Active Cruise Control)
15.ABS (Anti-Lock Braking System - DSC)
16.EPB (Electrical Parking Brake - EMF)/SRR_HL (Rear Radar Sensor Near Range Left)
17.PMA (Parking Assistant)
18.EPS (Electronic Power Steering)
19.TBX (Touchbox)
20.AMP (Amplifier)
21.EHC (Electronic Ride-Height Control)
22.CAS (Car Access System - CAS/FEM/BDC)
23.LHML (LED Main Light Module, Left)
24.LHMR (LED Main Light Module, Right)

25.EMA_LI (Left Electric Motor Reel)
26.EMA_RE (Right Electric Motor Reel)
27.FZD (Function Center, Roof)
28.KAFAS (Camera Based Driver Assist Systems)
29.GWS (Gear Selector Switch)
30.IC (Instrument Cluster - INSTR)
31.ECALL (Emergency Call)
32.CIC (Car Information Computer)
33.ZBE/CON (Controller)
34.FAS (Seat Module, Driver)
35.BFS (Seat Module, Passenger)
36.AHM (Trailer Module)
37.VDC/VDM (Vertical Dynamics Management)
38.AC (Air Conditioning - IHKA)
39.AC2 (Air Conditioning - IHKA2)

# All System Diagnostic Report

The Report is created by THINKCAR

## Repair Shop

Shop Name:Egeland Diagnostic
Address:Eigeland 50
Telephone:N/A
Sn:988569501368
Time:2026-01-17 16:27:08

## Vehicle Information

Year:2017.08
Make:BMW
Model:X5 xDrive40e iPerfor
VIN:WBAKT010XH0Y43663
Odometer:86610 miles
Vehicle Software Version:V10.30

## Inspection Result

SME (Battery Management Electronics) 1 problems exist

| | |
|---|---|
| 1.21F257 High-Voltage Battery, Cell Block: Increased Discharge Of A Cell | Intermittent |

HKFM/HKL (Automatic Luggage Compartment Lid Actuation) 1 problems exist

| | |
|---|---|
| 1.801724 HKFM, Automatic Operation Of Tailgate: Driving Speed Exceeds 0 km/h For Operating Requirement | Permanent |

ECM (Engine Control Module - DME/DDE) 4 problems exist

| | |
|---|---|
| 1.102A01 Air Mass Flow Sensor, Signal: Electrical Fault | Intermittent |
| 2.108C01 Charge Air Temperature, Plausibility: Temperature Too High | Intermittent |
| 3.120408 Charging Pressure Control: Switch-Off As Consequence | Intermittent |
| 4.10BA41 Outside Temperature Sensor, Plausibility: Ambient Temperature Less Than Model Temperature | Intermittent |

EME2 (Electrical Machine Electronics 2) 2 problems exist

| | |
|---|---|
| 1.2228C8 Starter Unit, Auxiliary Battery: Aged Auxiliary Battery | Permanent |
| 2.222884 Information Emergency Run Manager, DME: Request To Reduce The Output Of The Electric A/C Compressor | Intermittent |

IC (Instrument Cluster - INSTR) 1 problems exist

| | |
|---|---|
| 1.B7F6E3 HUD: Fault In Communication, HUD Does Not Respond | Intermittent |

ECALL (Emergency Call) 2 problems exist

| | |
|---|---|
| 1.031786 TCB: Functional Limitation Of Last State Call | Intermittent |
| 2.031789 TCB: Functional Limitation Of Remote Service | Permanent |

CIC (Car Information Computer) 2 problems exist

| | |
|---|---|
| 1.E1C440 HU-H: Reset | Intermittent |
| 2.E1C603 Headunit: Ethernet: Reset Of Communication | Intermittent |

AC (Air Conditioning - IHKA) 1 problems exist

| | |
|---|---|
| 1.80119D Stratification Flap Motor, Front Left: Locking Detected | Intermittent |

The following systems are OK:
1.SRS (Supplemental Inflatable Restraint System - AIRBAG/SGM-SIM)
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4.ZGM (Central Gateway Module)
5.LIM (Charging Interface Module)
6.EKPS (Electronic Fuel Pump)/RLM_1L (Tail Light Module, Outer Left)
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